Linear Regression

Python 3 (ipykernel)

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

In [2]:
```
data = {
    "Hours": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Score": [35, 40, 50, 55, 60, 65, 70, 78, 85, 92]
}

df = pd.DataFrame(data)
df
```

Out[ ]:

	Hours	Score
0	1	35
1	2	40
2	3	50
3	4	55
4	5	60
5	6	65
6	7	70
7	8	78
8	9	85
9	10	92

In [3]:
```
X = df[["Hours"]]
y = df["Score"]
```

In [4]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

X_train.shape, X_test.shape
```

Out[ ]:
```
((8, 1), (2, 1))
```

In [5]:
```
model = LinearRegression()
model.fit(X_train, y_train)
```

In [6]:
```
print("Intercept:", round(model.intercept_, 2))
print("Coefficient:", round(model.coef_[0], 2))
```

Out[ ]:
```
Intercept: 28.91
Coefficient: 6.31
```

In [7]:
```
y_pred = model.predict(X_test)

results = pd.DataFrame({
    "Hours": X_test["Hours"].values,
    "Actual Score": y_test.values,
    "Predicted Score": y_pred.round(2)
})

results
```

Out[ ]:

	Hours	Actual Score	Predicted Score
0	9	85	85.70
1	2	40	41.53

In [8]:
```
print("MAE:", round(mean_absolute_error(y_test, y_pred), 2))
print("MSE:", round(mean_squared_error(y_test, y_pred), 2))
print("R2 Score:", round(r2_score(y_test, y_pred), 3))
```

Out[ ]:
```
MAE: 1.62
MSE: 2.73
R2 Score: 0.996
```

In [9]:
```
new_data = pd.DataFrame({"Hours": [7.5]})
model.predict(new_data)
```

Out[ ]:
```
array([76.24])
```

In [10]:
```
plt.scatter(X["Hours"], y)
plt.plot(X["Hours"], model.predict(X), label="Regression line")
plt.xlabel("Hours")
plt.ylabel("Score")
plt.title("Linear Regression")
plt.legend()
plt.grid(True)
plt.show()
```

